

Friedman Test Results for VRP GA Component Configurations

This report reproduces the rank data and statistical summary contained in `vrp_friedman_test_results.csv`. The analysis compares 48 GA component configurations on 12 complete VRP benchmark instances. Lower ranks indicate better relative performance within an instance.

Statistic	Value	Interpretation
Friedman chi-square	540.709206	Large test statistic relative to df
p-value	4.105e-85	Strong evidence against equal performance of all configurations
Degrees of freedom	47	48 configurations - 1
Complete instances	12	Benchmark blocks used in the test
Configurations	48	GA component combinations compared

Best average ranks

Rank position	Configuration	Average rank
1	bs-ox-nom-bt	2.083
2	bs-ox-swpm-bt	2.083
3	bs-ox-insm-bt	2.167
4	bs-ox-nom-rw	4.167
5	bs-ox-swpm-rw	5.500
6	bs-ox-insm-rw	5.667
7	bs-pmx-swpm-bt	9.333
8	bs-pmx-insm-bt	9.917
9	bs-pmx-insm-rw	11.167
10	bs-pmx-nom-bt	12.167

Configuration key: st = Straight Split; bs = Bellman Split; pmx = partially mapped crossover; cx = cycle crossover; ox = ordered crossover; nox = no crossover; insm = insert mutation; swpm = swap mutation; nom = no mutation; bt = binary tournament; rw = roulette wheel.

Per-instance Friedman ranks - Straight Split configurations

Configuration	C1	C2	C3	C4	C5	C11	C12	G1	G5	G9	G13	G17	Avg.
st-pmx-inism-bt	12	22	15	19	21	21	19	23	16	26	25	19	19.83
st-pmx-inism-rw	10	23	20	18	17	19	22	20	21	17	24	24	19.58
st-pmx-swpm-bt	24	17	17	22	26	24	24	17	17	22	18	22	20.83
st-pmx-swpm-rw	19	20	21	25	18	9	18	21	23	16	26	25	20.08
st-pmx-nom-bt	14	21	22	26	28	26	21	18	24	14	13	23	20.83
st-pmx-nom-rw	27	26	25	21	24	22	25	22	20	20	19	28	23.25
st-cx-inism-bt	25	29	32	33	36	30	29	32	32	34	32	34	31.50
st-cx-inism-rw	26	32	31	31	30	25	33	29	28	32	33	33	30.25
st-cx-swpm-bt	32	33	36	34	33	31	35	31	33	29	34	32	32.75
st-cx-swpm-rw	28	30	29	28	31	29	30	27	27	28	30	30	28.92
st-cx-nom-bt	44	39	33	35	35	38	37	34	31	36	36	35	36.08
st-cx-nom-rw	43	44	30	36	34	40	38	36	36	35	35	36	36.92
st-ox-inism-bt	11	16	9	15	12	11	15	9	10	15	14	14	12.58
st-ox-inism-rw	17	14	16	10	16	18	20	19	12	18	20	21	16.75
st-ox-swpm-bt	8	11	11	16	15	10	16	12	6	13	17	17	12.67
st-ox-swpm-rw	20	18	12	20	20	20	17	24	15	24	21	16	18.92
st-ox-nom-bt	13	19	13	12	13	15	13	13	8	21	12	13	13.75
st-ox-nom-rw	18	15	14	13	19	17	14	8	13	23	15	15	15.33
st-nox-inism-bt	36	36	38	38	37	39	36	40	40	40	39	39	38.17
st-nox-inism-rw	40	37	40	41	41	44	40	41	39	41	40	41	40.42
st-nox-swpm-bt	38	40	42	43	43	41	41	43	42	43	41	43	41.67
st-nox-swpm-rw	42	41	41	40	44	37	43	42	38	38	42	37	40.42
st-nox-nom-bt	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.50
st-nox-nom-rw	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.5	47.50

Bold/shaded cells indicate rank 1 for the corresponding instance. The Avg. column is computed from the 12 instance ranks.

Per-instance Friedman ranks - Bellman Split configurations

Configuration	C1	C2	C3	C4	C5	C11	C12	G1	G5	G9	G13	G17	Avg.
bs-pmx-insm-bt	15	13	7	11	9	14	6	10	11	7	7	9	9.92
bs-pmx-insm-rw	16	10	8	9	8	13	8	14	18	12	10	8	11.17
bs-pmx-swpm-bt	9	8	19	8	7	7	12	11	7	4	9	11	9.33
bs-pmx-swpm-rw	23	9	10	17	11	12	11	15	22	10	16	7	13.58
bs-pmx-nom-bt	22	7	18	7	10	16	9	16	14	9	8	10	12.17
bs-pmx-nom-rw	7	12	23	14	14	8	10	7	19	11	11	12	12.33
bs-cx-insm-bt	30	27	28	30	23	28	28	25	29	30	27	18	26.92
bs-cx-insm-rw	21	25	26	27	29	27	27	28	26	25	22	26	25.75
bs-cx-swpm-bt	29	28	27	24	27	32	26	30	30	31	28	27	28.25
bs-cx-swpm-rw	31	24	24	23	25	23	23	26	25	19	23	20	23.83
bs-cx-nom-bt	37	38	35	32	32	35	31	33	34	27	31	31	33.00
bs-cx-nom-rw	34	31	34	29	22	33	32	35	35	33	29	29	31.33
bs-ox-insm-bt	1	2	2	4	1	3	2	1	1	2	4	3	2.17
bs-ox-insm-rw	5	5	6	5	5	6	5	3	9	8	6	5	5.67
bs-ox-swpm-bt	3	3	3	2	2	1	1	5	2	1	1	1	2.08
bs-ox-swpm-rw	6	4	5	6	6	5	7	6	5	5	5	6	5.50
bs-ox-nom-bt	2	1	1	1	3	2	3	2	3	3	2	2	2.08
bs-ox-nom-rw	4	6	4	3	4	4	4	4	4	6	3	4	4.17
bs-nox-insm-bt	35	35	37	37	38	34	34	38	37	37	38	40	36.67
bs-nox-insm-rw	33	34	39	39	40	36	39	37	41	42	37	38	37.92
bs-nox-swpm-bt	41	42	43	44	42	43	42	39	44	44	44	42	42.50
bs-nox-swpm-rw	39	43	44	42	39	42	44	44	43	39	43	44	42.17
bs-nox-nom-bt	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.50
bs-nox-nom-rw	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.5	45.50

Bold/shaded cells indicate rank 1 for the corresponding instance. The Avg. column is computed from the 12 instance ranks.

Average ranks for all 48 configurations

Pos.	Configuration	Avg. rank	Pos.	Configuration	Avg. rank
1	bs-ox-nom-bt	2.083	25	bs-cx-swpm-rw	23.833
2	bs-ox-swpm-bt	2.083	26	bs-cx-insm-rw	25.750
3	bs-ox-insm-bt	2.167	27	bs-cx-insm-bt	26.917
4	bs-ox-nom-rw	4.167	28	bs-cx-swpm-bt	28.250
5	bs-ox-swpm-rw	5.500	29	st-cx-swpm-rw	28.917
6	bs-ox-insm-rw	5.667	30	st-cx-insm-rw	30.250
7	bs-pmx-swpm-bt	9.333	31	bs-cx-nom-rw	31.333
8	bs-pmx-insm-bt	9.917	32	st-cx-insm-bt	31.500
9	bs-pmx-insm-rw	11.167	33	st-cx-swpm-bt	32.750
10	bs-pmx-nom-bt	12.167	34	bs-cx-nom-bt	33.000
11	bs-pmx-nom-rw	12.333	35	st-cx-nom-bt	36.083
12	st-ox-insm-bt	12.583	36	bs-nox-insm-bt	36.667
13	st-ox-swpm-bt	12.667	37	st-cx-nom-rw	36.917
14	bs-pmx-swpm-rw	13.583	38	bs-nox-insm-rw	37.917
15	st-ox-nom-bt	13.750	39	st-nox-insm-bt	38.167
16	st-ox-nom-rw	15.333	40	st-nox-insm-rw	40.417
17	st-ox-insm-rw	16.750	41	st-nox-swpm-rw	40.417
18	st-ox-swpm-rw	18.917	42	st-nox-swpm-bt	41.667
19	st-pmx-insm-rw	19.583	43	bs-nox-swpm-rw	42.167
20	st-pmx-insm-bt	19.833	44	bs-nox-swpm-bt	42.500
21	st-pmx-swpm-rw	20.083	45	bs-nox-nom-bt	45.500
22	st-pmx-swpm-bt	20.833	46	bs-nox-nom-rw	45.500
23	st-pmx-nom-bt	20.833	47	st-nox-nom-bt	47.500
24	st-pmx-nom-rw	23.250	48	st-nox-nom-rw	47.500

Statistical interpretation. The Friedman test rejects the null hypothesis that all 48 configurations perform equivalently ($\chi^2(47) = 540.709$, $p = 4.105e-85$). The best average ranks are obtained by **bs-ox-nom-bt** and **bs-ox-swpm-bt** (both 2.083), followed by **bs-ox-insm-bt** (2.167). All three combine Bellman Split, ordered crossover, and binary tournament selection, while differing in mutation choice. The omnibus Friedman result establishes an overall difference among configurations; post-hoc pairwise testing is required to determine which specific pairs differ significantly.